

Q1

Q2

Q3

Q4

Q5

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University of Canterbury

Term Test 2025

Prescription Number(s): **ENEL445**

Paper Title: **Applied Engineering Optimisation**

Time allowed: **1 hour 30 minutes**

Number of pages: **20**

Calculators **ARE** permitted.

ALL Questions ARE COMPULSORY.

(There are 5 questions in total.

The maximum total score is 100 marks.)

Family name (in block letters):

Given name:

Student number:

INSTRUCTIONS

This is an closed-book test (apart from the A4 cheat sheet). Please write you answers in the space provided. There is extra space at the end of the test should you need it. Show your working as you complete the test, so that you can more easily receive partial credit. Make a note indicating when your work is continued on an extra page.

Q1. Basics

(a) [6 marks]

Let $\mathbf{x}$ be a column vector of size $n \times 1$ and $\mathbf{A}$ be a matrix of size $n \times n$.

Show that the vector derivative of $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$ with respect to $\mathbf{x}$ is $\mathbf{A} \mathbf{x} + \mathbf{A}^T \mathbf{x}$.

[illegible]

Show that the infinity norm of an $n \times 1$ vector $\mathbf{x}$ is the maximum absolute value of the entries in $\mathbf{x}$.

[illegible]

Write down the Taylor series expansion around the point $\mathbf{c}$ for a multivariate function, $f(\mathbf{x})$, up to and including the second-order term.

Write down the definition of positive semi-definiteness and discuss the role of positive semi-definite matrices in the context of optimisation.

[illegible]

Q2. Unconstrained Optimization (1)**(a) [3 marks]**

Directional gradient: Substitute $\mathbf{x} = \mathbf{c} + \mu \mathbf{p}$ into your answer to Q1 (c), where μ is the step size, and keep up to the first-order term to derive the directional gradient of $f(\mathbf{x})$ along the direction $\mathbf{p}$.

(b) [5 marks]

First-order necessary optimality condition: Show that if $\mathbf{c}$ is a local minimum of $f(\mathbf{x})$, we must have $\nabla f(\mathbf{c})^T \mathbf{p} = 0$.

Hint: use the fact that the direction vector $\mathbf{p}$ can be arbitrary.

(c) (i) [2 marks]

Suppose the objective function is quadratic with respect to the design variable. That is, $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{A}\mathbf{x} + \mathbf{b}^T \mathbf{x}$, where $\mathbf{A}$ is positive definite. Compute its gradient at $\mathbf{x} = \mathbf{c}$ (i.e., find $\nabla f(\mathbf{c})$).

Hint: you may want to apply some of your answers to Q1 (a).

(ii) [4 marks]

Line search: We would like to perform an **exact** line search at $\mathbf{c}$ along the direction $\mathbf{p}$. Formulate this exact line search problem as an unconstrained optimization problem with the design variable being the step size μ .

(iii) [6 marks]

Optimal step size: Solve the unconstrained optimization problem formulated in the previous question for the optimal step size μ^* .

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Q3. Unconstrained Optimization (2)

Nonlinear least squares: Consider a range-based location system. We use three sensors at $\mathbf{s}_1 = [s_{x,1}, s_{y,1}]^T$, $\mathbf{s}_2 = [s_{x,2}, s_{y,2}]^T$ and $\mathbf{s}_3 = [s_{x,3}, s_{y,3}]^T$ to locate an object at unknown location $\mathbf{u} = [u_x, u_y]^T$.

(a) [5 marks]

Let $r_i(\mathbf{u}) = \|\mathbf{u} - \mathbf{s}_i\|_2$ be the **true** distance between sensor i and the object $\mathbf{u}$, where $i = 1, 2, 3$, and $\|\cdot\|_2$ is the Euclidean norm (2-norm).

Derive the gradient of $r_i(\mathbf{u})$.

(b) [5 marks]

Jacobian matrix: Derive the Jacobian $\mathbf{J}(\mathbf{u})$ of $\mathbf{r}(\mathbf{u}) = [r_1(\mathbf{u}), r_2(\mathbf{u}), r_3(\mathbf{u})]^T$.

Hint: use your result in Q3 (a).

(c) [10 marks]

Let d_i be the **measured** distance between sensor i and the object $\mathbf{u}$, where $i = 1, 2, 3$.

Formulate a nonlinear least squares problem with the design variable being $\mathbf{u}$. This is the problem to be solved for range-based location.

Hint: you can use the definitions $\mathbf{d} = [d_1, d_2, d_3]^T$ and $\mathbf{r}(\mathbf{u}) = [r_1(\mathbf{u}), r_2(\mathbf{u}), r_3(\mathbf{u})]^T$ to simplify your expression.

Q4. Constrained optimisation & algorithms

(a) [16 marks]

Consider the problem:

$$\begin{array}{ll} \min & (x_1 - x_2)^2 \\ \text{subject to} & x_1 + 2x_2 = 0 \end{array}$$

i) State the Lagrangian function [2 marks].

[illegible]

ii) Derive the KKT conditions associated with this problem and find the optimal solution. [6 marks].

[illegible]

[illegible]

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In a few sentences, explain the concept of *slackness* and slack variables.

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What is the "combinatorial problem" when trying to solve the KKT conditions of an optimisation problem? How does the Sequential Quadratic Programming algorithm (SQP) cope with this issue?

[illegible]

Q5. Optimisation-based Control**(a) [8 marks]**

Consider the system:

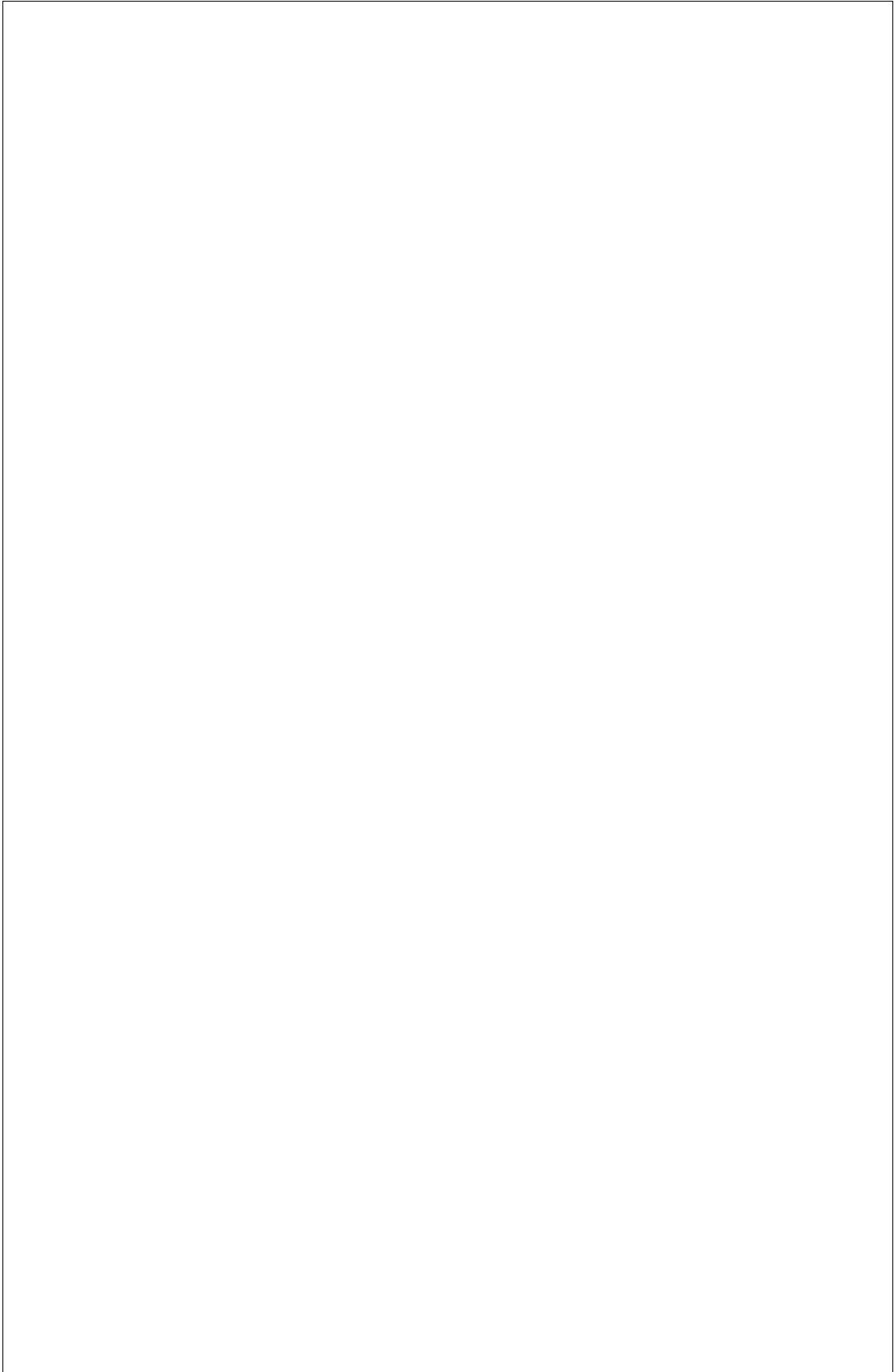
$$x_{k+1} = u_k$$

with cost function:

	$u = 1$	$u = 2$	$u = 3$
$c(x_k, u_k) =$			
$x = 1$	2	3	4
$x = 2$	1	2	3
$x = 3$	2	1	1

and terminal cost $J_h(x_h) = 1$. Find the optimal solution for two steps (i.e. the optimal u_{h-1} and u_{h-2}) for any initial value of $x_{h-2} \in \{1, 2, 3\}$ recursively using dynamic programming. (Hint: a graphical illustration might be helpful).

TURN OVER



Consider the discrete-time system:

$$y_k = x_k$$

i) Formulate (but do **not** solve) the model-predictive control problem [5 marks].

ii) What is the horizon length implied by the given reference trajectory [1 mark]?

iii) Is this (or can the problem be made) a quadratic optimisation problem [2 marks]?

Hint: you do not need to include terminal constraints / costs in your formulation (although you would not be penalised for doing so). However, you do need to make sure all the given information is represented in the problem.

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(c) [2 marks]

Explain briefly, ideally in two sentences or less, the trade-off when choosing ϵ in ϵ -greedy exploration, as often used with Q Learning.

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[illegible]

[illegible]

END OF PAPER